Integrating SAR and Optical Imagery Analysis for Liquefaction Phenomenon Identification of Post-Pohang Earthquake 2017, South Korea, Utilizing a Hybrid Deep-Learning Approach

Muhammad Fulki Fadhillah, Wahyu Luqmanul Hakim, Sung-Jae Park, and Chang-Wook Lee, *Senior Member, IEEE*

***Abstract*— An interesting liquefaction event happened following the 5.6-Mw Pohang earthquake on November 15, 2017. Liquefaction affects soil density as a result of earthquake vibrations, causing water to ascend and combine with solid soil. In general, this effect results from increasing water pressure in the buried lower layer. Remote sensing data, particularly those obtained with differential interferometry SAR (DInSAR), can be utilized to assess surface changes and soil moisture levels. This analysis utilizes Sentinel-1 C-band data from 2017 to 2020. Furthermore, the biased time-series identification presented to analyze the anomaly may be present in this study. In addition, optical satellite data were used to estimate changes in water content and soil moisture following Pohang earthquake. A combination of spectral bands sensitive to changes in water content was used to identify anomalies following an earthquake based on Sentinel-2 and Landsat-8 data. Based on the spatial analysis, optical images may detect changes in water content with a spatial accuracy of 60%–80% when compared to field data. Moreover, the liquefaction susceptibility map has been generated using a hybrid dense convolutional neural network (DCNN) architecture and swarm-based optimization algorithm. As a result, the susceptibility model performance was conducted using *k*-fold cross correlation with an area under the curve (AUC) value of 0.75–0.86. However, this research was the initial effort to determine the potential of liquefaction in the future based on the 2017 Pohang earthquake, and the results can improve our understanding of this phenomenon compression.**

***Index Terms*— Coherence, liquefaction, optical image, soil moisture, synthetic aperture radar (SAR).**

Received 10 July 2024; revised 27 November 2024 and 27 January 2025; accepted 8 March 2025. Date of publication 19 March 2025; date of current version 27 March 2025. This work was supported in part by the National Research Foundation of Korea provided by the Government of Korea under Grant 2019R1A2C1085686; in part by Korea Polar Research Institute (KOPRI) Grant funded by the Ministry of Oceans and Fisheries under KOPRI Project PE22900; and in part by the National Research Foundation of Korea (NRF) Grant funded by Korean Government (Ministry of Science and ICT), "Development of Application Support System for Satellite Information Big Data," under Grant RS-2022-00165154. *(Corresponding author: Chang-Wook Lee.)*

Muhammad Fulki Fadhillah and Wahyu Luqmanul Hakim are with the Department of Science Education and Interdisciplinary Program in Earth Environmental System Science and Engineering, Kangwon National University, Gangwon-do, Chuncheon-si 24341, South Korea (e-mail: fulkifadhillah@kangwon.ac.kr; wahyulhakim@kangwon.ac.kr).

Sung-Jae Park is with the Department of Smart Regional Innovation, Kangwon National University, Gangwon-do, Chuncheon-si 24341, South Korea (e-mail: tres13@kangwon.ac.kr).

Chang-Wook Lee is with the Department of Science Education and the Department of Smart Regional Innovation, Kangwon National University, Gangwon-do 24341, South Korea (e-mail: cwlee@kangwon.ac.kr).

Digital Object Identifier 10.1109/TGRS.2025.3550554

I. Introduction

EARTHQUAKES can cause secondary effects and nontectonic processes on the surface. One of these secondary effects is soil liquefaction due to cyclic loading. Liquefaction events generally occur in loose to moderately saturated granular soil and are dependent on the drainage properties of the soil [1], [2]. Due to the cyclic vibrations that occur at this time, the soil loses strength or stiffness to the point that it cannot stably support structures above it [3]. Pore water pressure increases because the moving soil causes water to flow upward; this changes the sand from a solid to liquid, resulting in liquefaction [4]. The earthquake in Pohang was the second-strongest earthquake ever recorded by Korean Meteorological Agency. Previously, a 5.8-Mw earthquake shook the city of Gyeongju, which remains the strongest earthquake ever recorded in South Korea [5]. Additionally, Pohang earthquake occurred in a soft soil geological area, which can increase the amplification of vibrations, causing extensive ground movement and creating the potential for liquefaction in the surrounding area [6].

Remote sensing has been increasingly used to investigate natural phenomena in recent years. Synthetic aperture radar (SAR) is a satellite-based active radar instrument used to detect changes in the Earth's surface [7], [8], [9]. The use of microwaves in the sensor is beneficial to the detection process because they penetrate the cloud and are not affected by light. This makes SAR a practical option for observing natural phenomena such as earthquakes and volcanic eruptions. Liquefaction has been extensively studied, with drilling data used to obtain the liquefaction potential index (LPI) in affected areas [10]. In addition, remote sensing has been used to detect liquefaction based on analysis of coherence values at the time the earthquake occurred [11], [12], [13], [14].

Previous studies have used the SAR approach to observe soil moisture at various wavelengths [15], [16]. Soil moisture measurement has also been advanced with the use of multispectral data using the PRISMA or OPTRAM approaches [17], [18], [19], [20]. However, one of the disadvantages of optical satellites is that optical satellites are quite sensitive to the atmospheric conditions around the observation area, and the image quality is reduced by cloud and water vapor. Previous studies have estimated deformation due to earthquakes using

© 2025 The Authors. This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

interferometry SAR (InSAR) data from Sentinel-1; these data were used to estimate the distribution of static slip on the fault plane due to Pohang earthquake, i.e., by reversing InSAR surface deformation data [21]. Coherence allows the detection of differences in consolidation and liquefaction damage at the building block scale and the detection of discarded sand and water [22]. The use of SAR data, especially amplitude and coherence data in time series, to identify soil moisture changes in peatlands [16], [23]. Also, the potential use of remote sensing and geotechnical data to improve the reliability of post-earthquake liquefaction detection [24]. In addition, the analysis of the closure phase based on InSAR data has attracted attention to observing changes in surface deformation [25]. Despite the developments, there is still room for improvisation and increasing reliability in liquefaction detection using remote sensing data. More specifically, there has been no research on the use of SAR data to determine the liquefaction that occurred after the 2017 Pohang earthquake. With various approaches in the use of optical data for soil moisture changes and their potential use in liquefaction identification. In addition, the potential use of InSAR phase data in the form of time-series bias in monitoring soil moisture changes so that it can provide comprehensive information based on SAR and optical data in liquefaction identification as well as providing insight into the potential for liquefaction susceptibility mapping based on multiremote sensing data.

Our study aims to understand this challenging phenomenon using the usefulness of radar and optical satellite data by investigating and analyzing the temporal effects of liquefaction and demonstrating consistency with the DInSAR observations (i.e., surface deformation). In addition, the coherence values in interferometry pairs were also evaluated to understand the difference in coherence throughout the year associated with precipitation, using observation data from 2016 to 2020 derived from Korea Meteorological Agency (KMA). Therefore, we also provide improved combined scatterers interferometry with optimized point scatterers (ICOPS). A combination of the distributed scatterer (DS) points with persistent scatterer (PS) points can provide better spatial coverage and measurement. In addition, the optical satellite we derived from the Landsat-8 and Sentinel-2 in the pre- and co-seismic events will be analyzed for the change of moisture post-earthquake. Satellite imagery of the liquefaction phenomenon was then compared to existing field data to validate the results. Moreover, we also generated a liquefaction susceptibility map from the satellite imagery data by applying the dense convolutional neural network (DCNN) architecture algorithm after Pohang earthquake in 2017. For the susceptibility validation process, we evaluated the performance of the generated model using the area under the curve (AUC) value defined from the receiving operating curve (ROC) with k-fold cross validation [9], [26]. However, this is the first attempt to generate a liquefaction susceptibility map in Pohang induced by earthquakes using radar and optical imagery data. This approach represented the initial attempt to use SAR interferometry data of the liquefaction occurring after the 2017 Pohang earthquake.

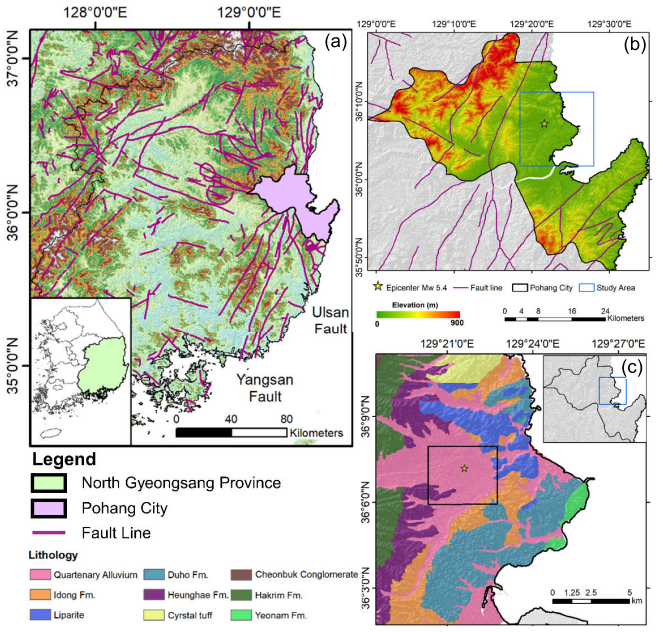


Fig. 1. Location of the study area. (a) Map of South Korea with the red shape representing Pohang administration boundaries. (b) Map of Pohang district with elevation data with a purple line representing the fault system in Pohang district and the yellow star representing the epicenter of Pohang earthquake. (c) Lithology condition of the study area.

This research was organized as follows. Section II presents the background of the study area and data used in this research. Then, Section III describes the methodology of processing SAR and optical data for liquefaction analysis in post-seismic events, and then, the proposed liquefaction susceptibility map based on a deep-learning algorithm. Then, the identification of coherence, surface deformation with bias time-series analysis, and post-event analysis based on optical data was presented in Section IV, including a liquefaction susceptibility map. In Section V, the validation assessment was conducted for optical imagery, and a performance analysis of susceptibility was provided. Section VI presents the conclusion and future direction of the research.

II. Data and Study Area

A. Study Area

Pohang is located in North Gyeongsang in the east of South Korea, close to the East Sea [see Fig. 1(a)]. Pohang has an area of 1127 km^2 and a population of 516 471, according to the 2010 census. This area is hilly in the northeast and includes a beach composed of semi-consolidated tertiary sediments known as Pohang Basin [27]. The northwest of Pohang has a mountainous topography composed of volcanic rocks and granite, which are common in Korean Peninsula, as shown in Fig. 1(b) and (c). Pohang has a semi-tropical climate, with an average annual rainfall of 1119 mm from July to September. In terms of its geological characteristics, the study area is mainly composed of limestone, granite, and volcanic sedimentary rocks [28]. In the study area, Cretaceous volcanic rocks and granite are distributed mainly in the west, while tertiary deposits are distributed in the east, and tertiary volcanic rocks

TABLE I
SPECIFICATIONS OF THE SAR DATA SATELLITES

NO	Satellite	Radar band	Direction	Temporal coverage	Number of scenes
1	Sentinel-1	C-Band (5.5 cm)	Descending	March 2017 – March 2020	104

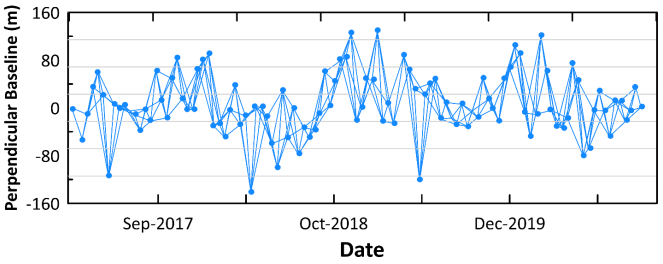


Fig. 2. Baseline perpendicular of Sentinel-1 dataset with maximum temporal day 60 days and baseline perpendicular 160 m.

and granite are distributed in the south, as shown in Fig. 1(c) [29].

In November 2017, Pohang earthquake (5.6 Mw) damaged the surrounding area. In addition, ground liquefaction was detected close to the epicenter of the earthquake. This is a post-earthquake effect, in which the soil conditions become more fluid due to water intrusion into the underground layers exposed to the earthquake [10]. This study focuses on events related to the liquefaction phenomenon after Pohang earthquake.

B. SAR Data Processing

To construct an interferogram, we mainly used data from the area affected by the earthquake. Prior to the interferogram formation process, all SAR data in the single-look complex (SLC) format were used in the co-registration process. In this study, we used the multiband SAR data available from the Sentinel-1 C band (2016–2020), as presented in Table I. A multimaster scheme was used to construct an interferometry time series for the analysis of changes in the deformation of multitemporal surfaces.

However, SAR data from the Sentinel-1 C-band, which has a short repetition cycle, have advantages from a temporal aspect, reducing the possibility of temporal decorrelation and promoting coherence. Therefore, for the InSAR time-series deformation analysis after Pohang earthquake, we used an ICOPS analysis technique [30]. After performing the generating process using a multimaster interferogram, we selected an interferogram with good results, a temporal delay of 60 days, and a maximum perpendicular baseline of 160 m, as shown in Fig. 2. This method also provided analytical advantages by utilizing an iteration process that reduced the calculation errors that occurred during the previous process [31].

III. METHOD

This study uses several sequences to analyze the liquefaction phenomenon after Pohang earthquake of 2017 using time-series InSAR, optical imagery, and a liquefaction susceptibility map. For simplification, this study can be separated into three steps which are the SAR process, the optical imagery process, and the liquefaction susceptibility map. The workflow of this study can be shown in Fig. 3.

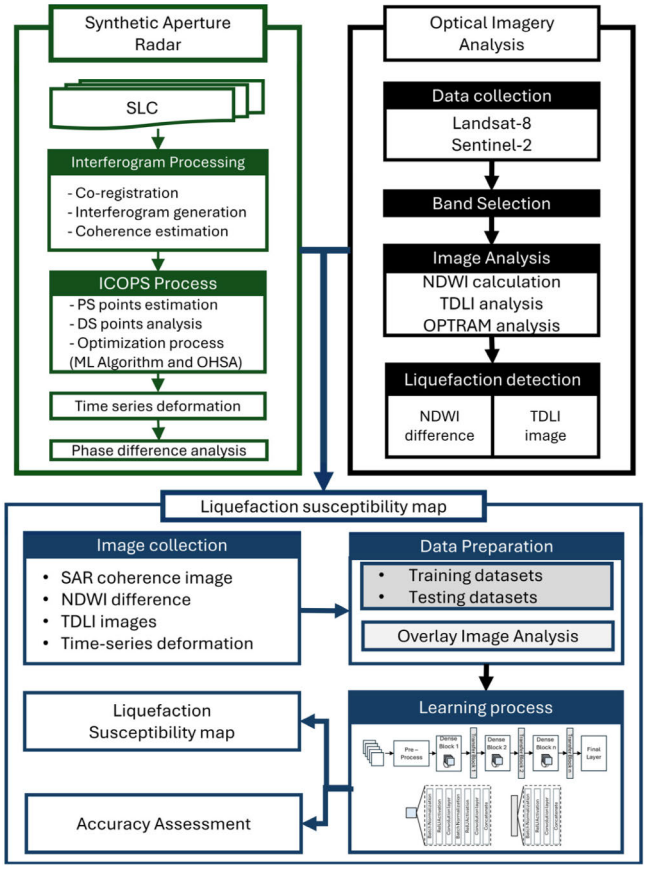


Fig. 3. General workflow of the research that processed the radar data to generate time-series InSAR and optical imagery data analysis for liquefaction.

A. Interferometric Processing

After the SLC sentinel-1 data acquisition process in the 2016–2020 timeframe, we will start the process with SLC data co-registration. In general, co-registration begins by selecting data in the middle of the time distribution, which will act as a reference image. In order to refine the SLC data, orbital file data are applied to provide corrections to minimize errors in the next process. In the process of forming the interferogram, the results of the registration image will be used by considering consecutive date pairs by taking into account the temporal distance of the data. To extract the differential interferogram SAR (DInSAR), we used a topographic phase based on Copernicus digital elevation model (DEM) with a resolution of 30 m, which resulted in an interferogram with a dominant deformation phase and an error phase (atmospheric artifact and noise). After obtaining the DInSAR, the geocoding process is also carried out to facilitate the analysis process using the coordinates contained in the DEM. On the other hand, the changes between two SAR data acquisitions that occur due to changes in the angle of incidence of the beam and noise disturbance can serve as indicators in observations which,

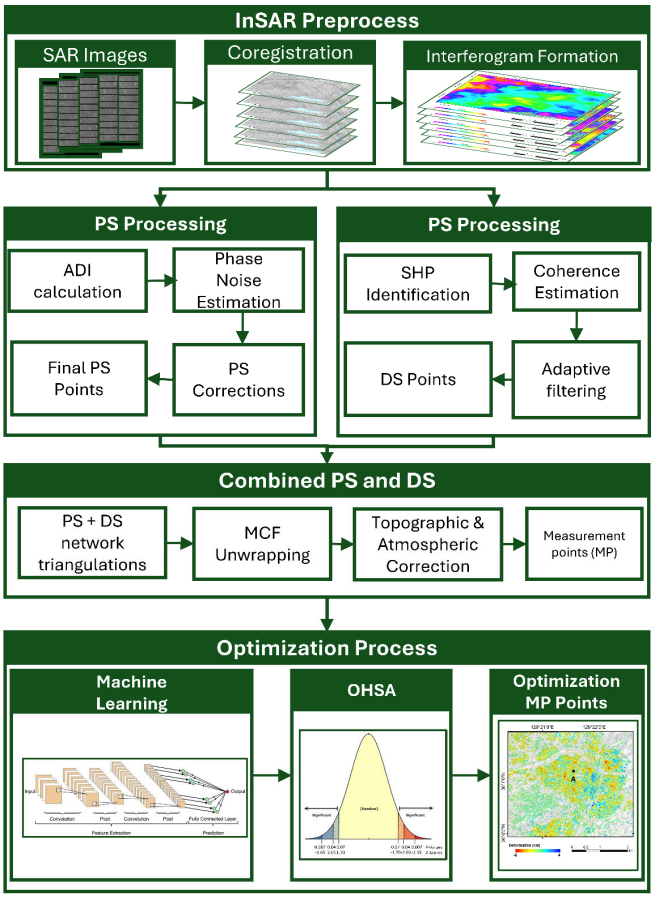


Fig. 4. Flowchart of the ICOPS process which was conducted from the interferometric process, PS and DS identification, and optimization process.

called coherence [32]. For the coherence pixels, the window slides from pixel to pixel. A weighting function (linear or Gaussian) can be applied within the window to reduce the loss of resolution due to windowing operations [33].

B. ICOPS Analysis

The ICOPS process begins with estimating PS points and DS points before analyzing the time-series deformation based on the combination of PS and DS [30]. The combination of DS and PS is expected to maximize the distribution of measurement point (MP) in spatial density and produce more reliable deformation results, and the flowchart of the ICOPS process is shown in Fig. 4. In general, the determination of PS points is based on the amplitude dispersion index (ADI) analysis [34]. Estimated PS points candidates are expected to select points that have less effect than decorrelation noise, which makes them reliable. The PS process sequence begins with an estimation of the phase noise in each interferogram pair in order to get PS candidates. Then, PS candidates will be re-estimated based on the characteristics of the phase and amplitude dispersion in order to get a reliable PS point. The selected PS points will be combined with the DS points to form a new network that will be used in the time-series deformation analysis process.

After determining the PS points, the DS point measurement is generally done by calculating the cluster of statistically homogonous pixels (SHPs) and estimating the coherence of the candidate DS points. In this study, the general likelihood ratio test (GLRT) was used to analyze adjacent SHP clusters in the SAR data [35]. This GLRT method provides a faster processing time and is able to produce reliable data in less data. The maximum likelihood estimate for the Rayleigh parameter of the dataset is expressed in the following equation:

$$\hat{\sigma}^2 = \frac{1}{M}\sum_{i=1}^{M} x_i^2, x_i = \text{sample data} \tag{1}$$

with the hypothesis assumption described in the following equations:

$$\mathrm{H}_0 : \hat{\sigma}^2 = \hat{\sigma}_p^2 = \hat{\sigma}_q^2 \tag{2}$$

$$H_1 : \hat{\sigma}_p^2 \neq \hat{\sigma}_q^2. \tag{3}$$

The estimation of SHPs was based on the similarity of amplitude $\hat{\sigma}$ values over time in (2) and (3), in which H_0 was the correct hypothesis assumption. The coherence complex value was also analyzed on the interferogram pair, which will be used as a reference for determining the DS point.

In determining the DS points, initially, we selected parameters to ensure the reliability of the DS points, starting from determining the SHP value >20, the mean spatial coherence more than 0.2, and we considered the temporal coherence >0.6 in determining the DS processing [36]. After the identification, the deformation analysis process will be carried out on the time-series interferogram stack. PS and DS pixels will be synchronized with the minimum cost flow (MCF) algorithm in the deformation phase analysis process. A singular variable decomposition (SVD) is needed as a solution for calculating the level of surface deformation in a time series. In addition, topographic phase correction is also carried out in the DInSAR phase to reduce the effect of topographic errors [37], [38]. In addition, low-pass filtering and high-pass filtering are also applied to calculate residual noise including atmospheric artifacts and orbital errors.

In this postprocess phase, we propose a machine-learning optimization approach to evaluate MPs obtained in a previous step and determine the appropriate parameters for a convolutional neural network (CNN) algorithm [8]. We evaluate all MPs to obtain linear coefficient values associated with deformation and correlation coefficients across these datasets. We perform point selection by excluding MPs with a linear correlation coefficient below 0.75 and group the remaining points based on their correlation coefficient. Points with a correlation coefficient greater than 0.7 are labeled "1," while the others are labeled "0." For the learning process purpose, the datasets were split into training and testing sets with a 70:30 ratio and used the CNN algorithm to generate an optimum map. The optimum MPs are selected based on the statistical properties of the deformation model, with the aim of reducing the number of points that have a low impact and are associated with noise and error. In addition to machine learning, the optimization hotspot analysis (OHSA) method was applied to the MP datasets for spatial clustering analysis. We assess and aggregate the data to ensure that the OHSA has an adequate number and variety of datasets and determine the

scale for the Getis-Ord Gi* analysis using incremental spatial autocorrelation [39], [40]. This approach minimizes external intervention by focusing on the statistical properties of the data. Clustered MP can provide a more reliable deformation map based on the topographic and spatial characteristics of the clustered MP.

C. Optical Image Analysis

Liquefaction is a phenomenon caused by changes in moisture conditions in the soil after an earthquake. This can occur when the density of water molecules in the soil increases, causing rigid particles to mix with water and affecting the soil moisture content. One way to analyze changes in soil moisture is to use the normalized difference water index (NDWI), which has been widely used to analyze moisture in vegetation and water bodies [41]. By analyzing changes in water content and soil moisture, it is possible to assess the likelihood of liquefaction. In addition, it is also possible to use an optical satellite-based temporal difference liquefaction index (TDLI) for liquefaction analysis by using a single SWIR optical band, which has been shown to be effective at detecting changes in soil moisture [42]. This method can complement NDWI-based analyses, which are useful for assessing the surrounding vegetation. Furthermore, we introduce a technique for computing soil moisture using the OPTRAM model in addition to the aforementioned methods. The purpose of this model is to measure the amount of moisture in the soil by calculating the area of the pixel distribution between the normalized difference vegetation index (NDVI) and the SWIR transformed reflectance (STIR) using a trapezoidal method [43]. By determining the dry edge and wet edge parameters, as indicated in (4) and (5), one may acquire the normalized SM data (W) for each pixel [18]. This is achieved by identifying the boundary where moisture is absent

$$W = \frac{(\mathrm{STR} - \mathrm{STR_d})}{(\mathrm{STR_w} - \mathrm{STR_d})} \tag{4}$$

$$\mathrm{STR} = \frac{(1 - R_{\mathrm{SWIR}})^2}{2R_{\mathrm{SWIR}}} \tag{5}$$

where $\mathrm{STR_d}$ and $\mathrm{STR_w}$ represent STR in dry and wet conditions, respectively [44].

We used optical satellite data from the Landsat-8 and Sentinel-2 satellites to assess liquefaction before and after an earthquake. For Landsat-8, we used bands 3, 5, and 6 for green, NIR, and SWIR, respectively. For Sentinel-2, we used bands 3, 8a, and 11 for green, NIR, and SWIR, respectively [17]. We calculated NDWI values for the pre- and post-earthquake periods to identify areas affected by liquefaction. However, we were unsure whether using sensitive and insensitive bands would be more effective at detecting surface changes due to liquefaction. Additionally, we calculated TDLI values by comparing pre- and post-earthquake periods and compared them to the time-series analysis method based on the difference between the three NDWI values. A positive TDLI value indicates higher water content and a higher likelihood of liquefaction, while a negative TDLI value indicates lower water content and a lower likelihood of liquefaction. We adjusted and calibrated the optical satellite data to account for radiation and atmospheric effects and co-registered the images to reduce geometric errors in the analysis of post-earthquake liquefaction location [17].

D. Liquefaction Susceptibility Map

1) Dense CNN: Decreasing the speed of handling large data faced by the conventional artificial neural network has been issued in the deep-learning development [45]. The variance of type data and development of the amount of data has triggered the researcher to overcome the limitations of the conventional approach, and then, it proposed the CNN. The convolutional layer includes several filters and layers from convolutional layers, pooling layers, and fully connected layers that divide the input data into smaller dimensional parts with simultaneous information collection with feature extraction [46], [47]. CNN's DenseNet architecture is a network architecture that was created to address the issue of residual network-based network degradation. The advancement of DCNN is complemented by enhanced characteristics in every layer, with the DCNN network structure consisting of numerous dense block and transition layers, as shown in Fig. 5(a). The dense block is a component of the model that includes many convolution layers, along with batch normalization, ReLU activation, and 1-D convolution [48]. The transition block comprises convolution and pooling layers that try to decrease the dimensionality of the data. Additionally, there is a convolution block that comprises a more complex convolution layer and batch normalization. The architectural model starts with the initial convolution layer and then proceeds with the dense block and transition block [49]. The last component of the model is implemented using global pooling and dense blocks, which are complemented by sigmoid activation.

2) Optimization Process: In this study, utilization of optimization techniques is employed to minimize the need for intervention and trial-and-error assumptions when utilizing the model. The optimization algorithm seeks to enhance the dependability of the model by optimizing the hyperparameters in the DCNN model. Swarm-based optimization algorithms, such as the gray wolf optimization (GWO) and particle swarm optimization (PSO), are used to make this optimization algorithm perform [8], [50]. GWO and PSO are optimization methods that mimic natural behavior and have proven effective in various optimization applications, including disaster vulnerability mapping [51]. GWO optimizes the neural network architecture, adjusting the number of layers and neurons to minimize prediction errors and improve model generalization [52]. Meanwhile, PSO focuses on optimizing the network weights and biases, accelerating convergence to the optimal solution, and avoiding local minima [53]. The integration of GWO and PSO creates a dynamic framework, which allows for adjustment of hyperparameters such as learning rate and momentum based on model performance. This adaptation also responds to changes in inputs to improve model reliability and adaptability. The optimization algorithm results in this research will regulate DCNN architecture hyperparameters, including the number of blocks, filters, batch size, and learning rate. The workflow of the liquefaction susceptibility map can be shown in Fig. 5(b).

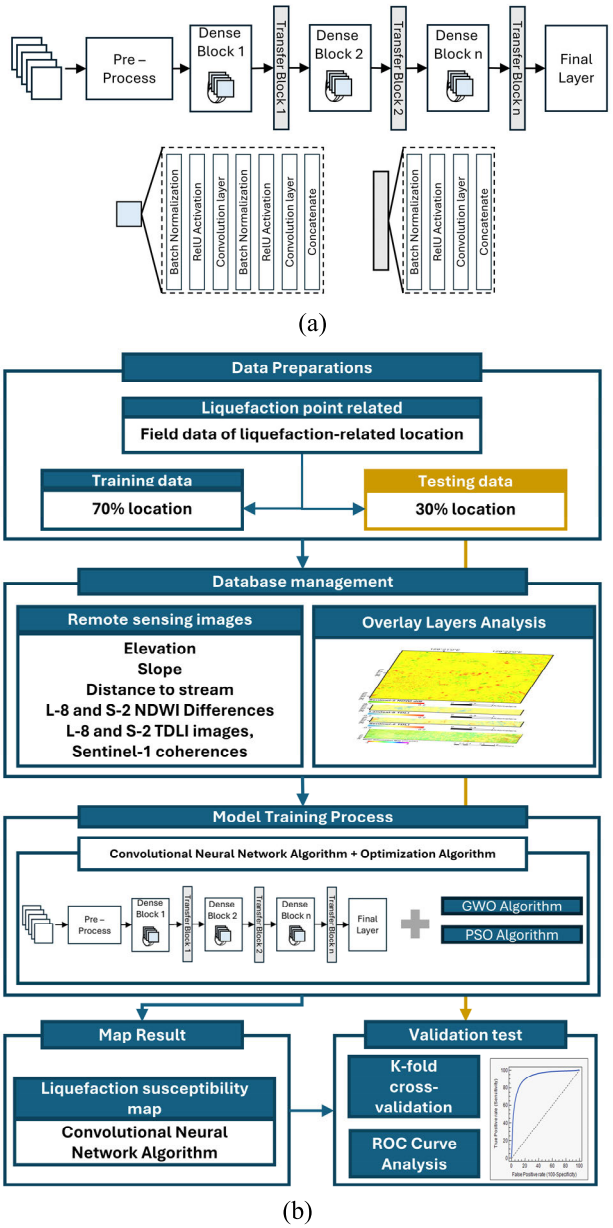


Fig. 5. (a) DCNN architecture illustration and (b) liquefaction susceptibility map workflow process using the Hybrid-DCNN algorithm.

For the influential factor, we use satellite imagery data from optical data derived from the Sentinel-2, and Landsat-8 has been chosen as the input imagery. The optical data have been processed for analyzing the liquefaction using the NDWI differences and the TDL index. Also, the radar satellite imagery provided by the Sentinel-1 C data has been integrated as the input for the susceptibility analysis. Coherence from the complex image and the time-series surface deformation map, which provide the information, has been evaluated to assess the liquefaction susceptibility in the paddy area near the epicenter [54], [55]. Besides, the elevation, slope, and topographic wetness index (TWI) were derived from the digital elevation model used in the model constructions. The distance to the stream was considered to be an influential factor in this study, and detailed information on spatial factors is shown in Table II. All the remote sensing imagery has been managed and handled

TABLE II
INFLUENTIAL LIQUEFACTION SPATIAL FACTOR USED IN THIS RESEARCH

Categorial Factors	Spatial Factors	Source
Topographical factor	Elevation	Copernicus Digital Elevation Model, European Space Agency (ESA)
	Slope	
	Topographic wetness index	
Geological Factors	Distance to fault	Korean Institute of Geoscience and Mineral (KIGAM)
	Lithology	Korean Institute of Geoscience and Mineral (KIGAM)
Environmental Factors	Distance to stream	National Geographic Information Institute (NGII)
	Modified Normalize Different Wetness Index	Landsat-8 Sentinel-2
	Temporal Different Liquefaction Index	Landsat-8 Sentinel-2
	NDWI Difference	Landsat-8 Sentinel-2
	Average Spatial Coherence	Sentinel-1
	Surface deformation	Sentinel-1

using the overlay analysis of all data. For the inventory, a database has been provided from the aerial and field data survey, which contains the liquefaction-related location around the study area [10], [17]. Then, the liquefaction susceptibility map was determined by the learning process using the CNN deep-learning algorithm. As the final result, the validation was carried out based on the ROC analysis to determine the AUC value. Besides, evaluation metrics, including mean square error (mse) and root mse (RMSE), were calculated [26]. Those statistical measurements represent the differences in predicted results and reference values, which are commonly used for regression analysis.

IV. RESULTS

A. Analysis of the Relationship Between Coherence and Precipitation

According to the results of the coherence analysis of the interferogram generated from the Sentinel-1 satellite data collected from 2017 to 2020, there was a seasonal trend in the coherence value in paddy fields and open land located around the earthquake's epicenter. The lowest coherence value of the interferogram pair occurred in the summer of every year. The coherence value in the interferogram pair is extracted from point A in the paddy area, as shown in Fig. 6(a). Also, the coherence in different time acquisition was analyzed in the paddy area near the epicenter of the earthquake, which shows the highest coherence value being recorded in spring. Seasonal fluctuation in coherence was reflected in a "V" pattern in the observation period 2017–2020. Precipitation data in the

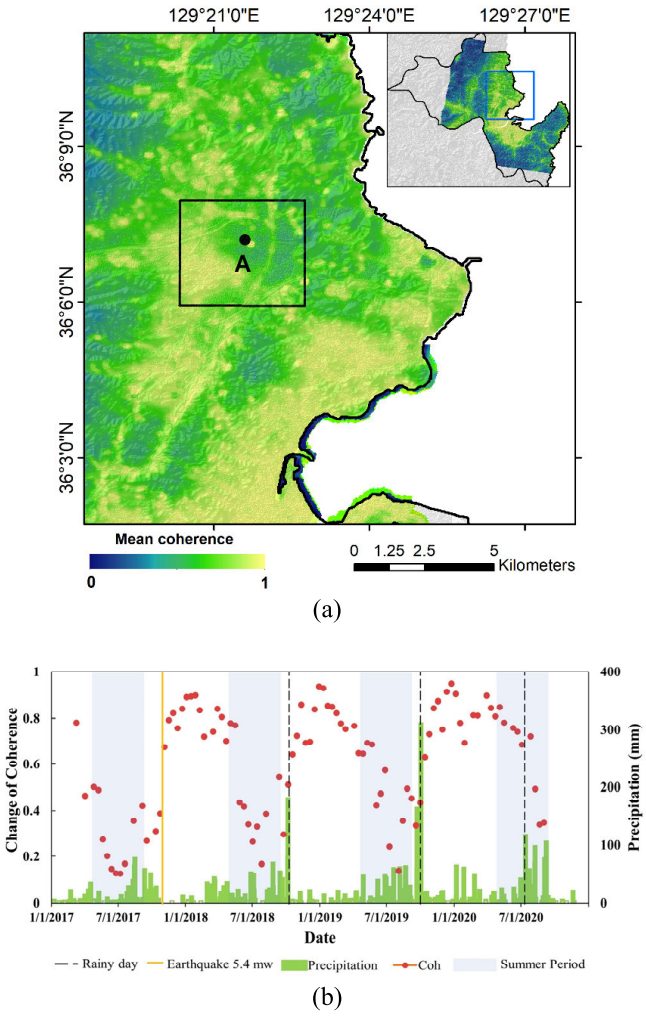


Fig. 6. (a) Mean coherence of the Sentinel-1 images in the study area (Pohang) from 2017 to 2020 with the image represented in the blue box in the inset image on the right corner and (b) change of coherence and precipitation graph generated from Sentinel-1 data for 2017–2020.

study area were collected for comparison with coherence data to analyze the effect of changes in precipitation under interferometric coherence. Fig. 6(b) shows the relationship between the precipitation data and coherence value. When the precipitation was high enough (>100 mm/day), there was an increase in interferometric coherence, as indicated by the blue dashed line in Fig. 6(b). The coherence values increased mostly in months with high rainfall, especially late summer; however, there was no increase in coherence after high rainfall. This is shown by the orange line in Fig. 6(b), which indicates a large change in coherence at the time of the earthquake. As high precipitation was not recorded in the period, the increase in coherence indicated a change in the soil moisture content in this area. This indicated the potential for liquefaction, which was reported to have occurred after the earthquake in Pohang, especially in rice fields and bare land.

B. Optical Satellite Imagery Identification

Optical imagery analysis has been generated from the Landsat-8 and Sentinel-2 satellites using the NDWI and TDLI methods. In this study, we have evaluated the NDWI differences using imagery from Landsat-8 and Sentinel-2 satellites. Also, the TDLI analysis has been carried out using satellite data from the SWIR band, which has a sensitivity to detecting the change of soil moisture. Therefore, here we show the example of the NDWI differences from Landsat-8 and TDLI image from Sentinel-2 satellite in Fig. 7(a) and (b), respectively. Moreover, the detailed analysis will be delivered in the following.

The NDWI results showed that the index increased significantly in the sandblast closest to the epicenter in Fig. 7(c) and (d). This indicates that this area was most severely affected by earthquake liquefaction. On the other hand, the index value did not significantly change in other sandblast areas. This indicates that either these areas were not severely affected by liquefaction or the NDWI analysis could not detect earthquake liquefaction in the area. This indicates that the NDWI approach had limitations in detecting surface liquefaction caused by an earthquake. Fig. 7(e) and (f) shows the results of TDLI mapping based on the SWIR band from Landsat-8 and Sentinel-2. The higher the index value (red pixels), the higher the water content on the surface. The Landsat-8 image had a high index value, with the red color indicating post-earthquake liquefaction. The TDLI value was higher than that obtained by the NDWI method. This indicates that the use of the SWIR band, which is more sensitive to changes in water content, would be appropriate. Fig. 7(f) shows the Sentinel 2 TDLI values in the liquefaction region; there were more obvious differences in the TDLI than NDWI according to Sentinel-2 imagery. For the NDWI values showing the seasonal characteristics throughout the year, this is suitable with the condition of the landscape that reflects the signal. Compared with the accumulation of seven-day precipitation, the seasonal characteristic NDWI is strongly related to the accumulation of precipitation, which increases during the rainy season and then decreases during winter [56]. Besides, the response of NDWI was delayed after the precipitation accumulated. During the earthquake, there is a little anomaly of NDWI characteristics, which may related to the liquefaction process. Besides, we acquired the soil moisture estimation from the OPTRAM approach from 2017 to 2020 from Landsat-8, as shown in Fig. 7(g), with the precipitation data from 2017 to 2018. Based on the result, the pattern of soil moisture presents a seasonal pattern, with the lowest value occurring during the summer session. During rainy days, the soil response is delayed, increasing the signal from the graph, and a fluctuation signal appears due to the cloud or atmospheric condition. Shortly after the earthquake happened, the signal showed the increasing signal in point A that reported sand bowl liquefaction appeared [44]. This finding can be notable, as the atmospheric condition influences the signal. However, soil condition and interference with atmospheric conditions may affect the time-series result.

C. InSAR Deformation Analysis

Fig. 8 represents the mean deformation map derived from the Sentinel-1 C-band satellite with the ICOPS method

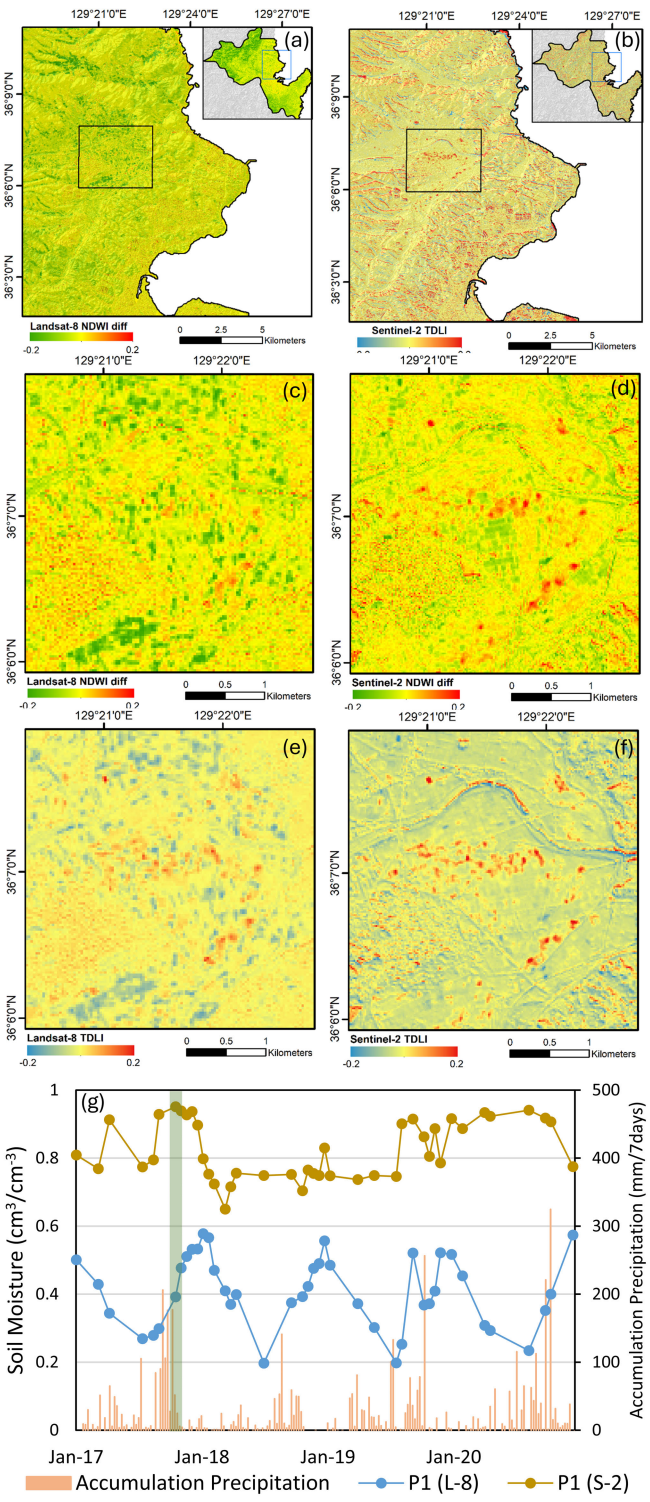


Fig. 7. (a) Landsat-8 NDWI differences and (b) Sentinel-2 TDLI in Pohang City during the co-seismic event. Also, zoomed-in view from of NDWI results obtained using (c) Landsat-8 and (d) Sentinel-2 imagery, and TDLI results obtained using (e) Landsat-8 and (f) Sentinel-2 imagery with (g) graph of OPTRAM results from Landsat-8 and Sentinel-2 with green line represent the event of Pohang earthquake 2017.

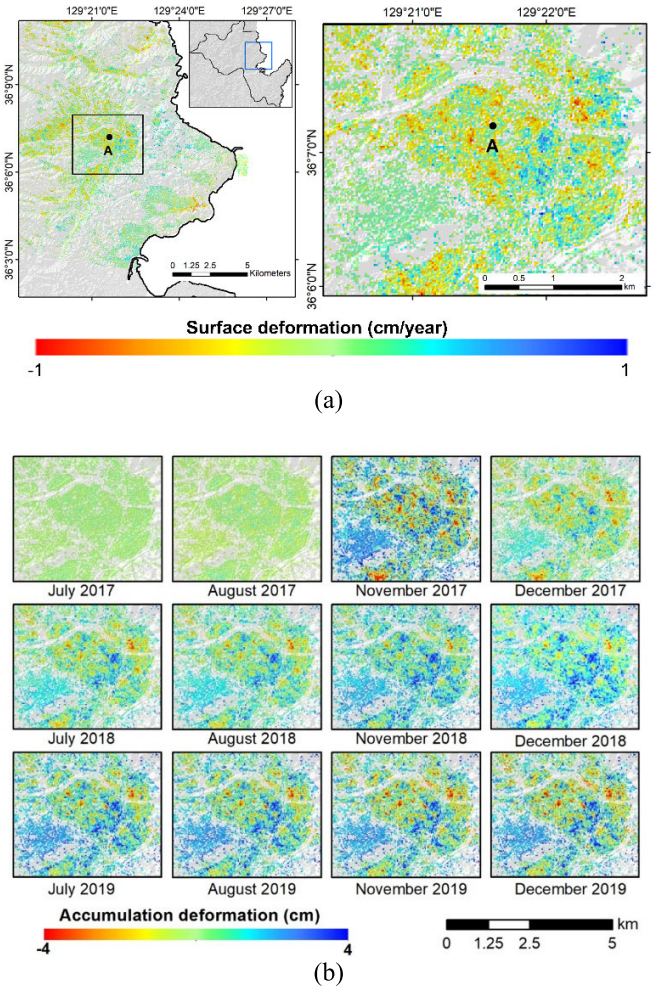


Fig. 8. (a) Mean deformation map generated from Sentinel-1 data for Pohang (2017–2020) the zoomed-in view of closed box with point A and (b) time-series map accumulation displacement in Pohang with November 2017 is the co-seismic event.

from 2017 to 2020. According to a time series (2017–2020) of InSAR observations, there was no change in deformation other than in areas affected by the earthquake in November 2017, as shown in Fig. 8(a) and (b). After the earthquake, there was a change in deformation around the earthquake epicenter, with a land subsidence of about 4 cm. These deformation changes were distributed to the northwest of the earthquake epicenter, with a reasonably wide spatial distribution. The distribution of deformations was likely a consequence of the soil characteristics around the epicenter, with quaternary alluvium amplifying vibrations and showing susceptibility to deformation. This allowed for local deformation of the post-earthquake fringe, but because of the low coherence level, local deformation was not significant.

While we identified a relationship between phase change and precipitation based on InSAR time-series results, soil moisture and vegetation have been proposed as two possible causes of the observed bias in the time series [55], [57], [58]. For the bias analysis, we conducted the time-series bias analysis of bandwidth-1 (bw-1) and bandwidth-3 (bw-3) for this study. Fig. 9(a) and (b) represents the mean velocity map from bw-1 and bw-3, respectively. As the result of the mean bias velocity map, it presents dominance of subsidence in this paddy area of subset images [Fig. 9(a) and (b)]. The signal shows a consistent area in bw-1 and bw-3, which may relate to the agricultural activity in this area. Meanwhile, noticeably, it appears to uplift the signal with no significant amount in the

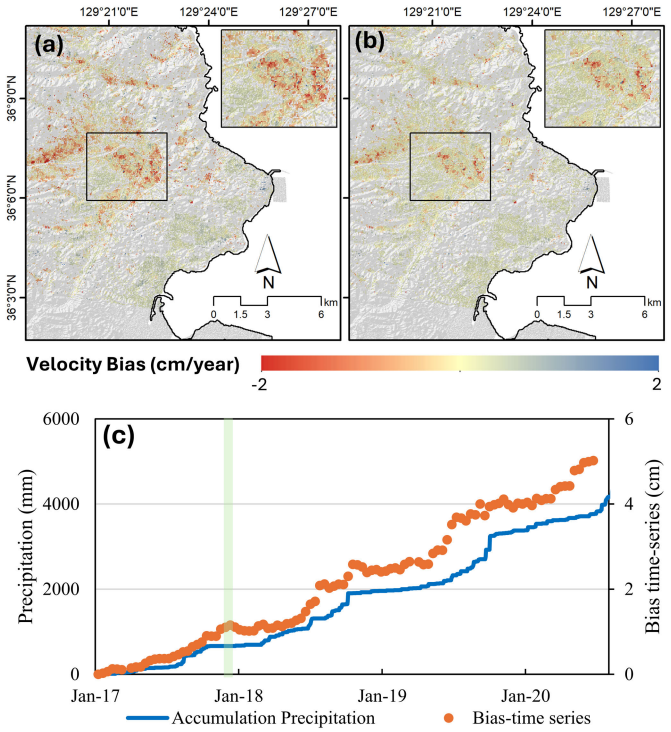


Fig. 9. Bias of velocity maps from (a) bw-1 and (b) bw-3. (c) Bias time-series graph of bw-1 with accumulation precipitation data is marked in Fig. 8 and a green line indicates Pohang earthquake in 2017.

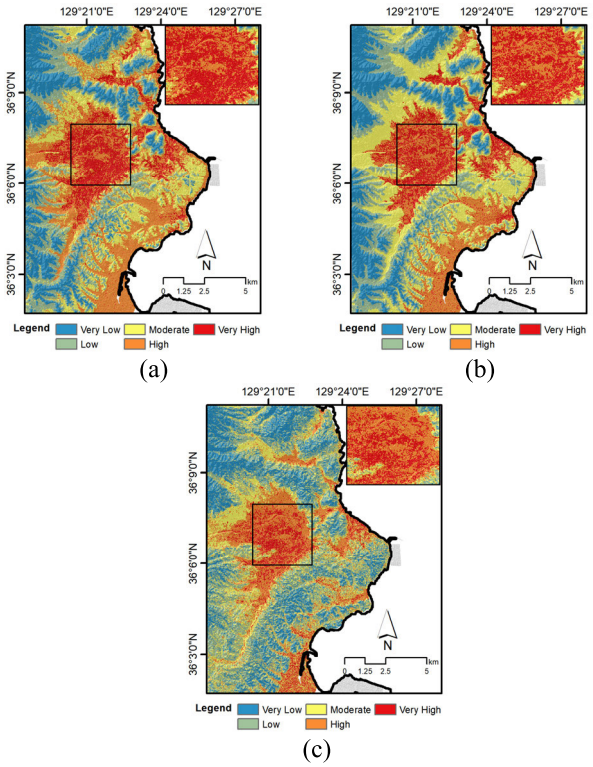


Fig. 10. DCNN algorithm liquefaction susceptibility map results from (a) DCNN-base, (b) DCNN-GWO, and (c) DCNN-PSO.

paddy area in either bandwidth. We examined areas showing significant deformation in the rice fields near the earthquake epicenter, indicating liquefaction. We obtained precipitation data from the nearest station from KMA and compared the cumulative precipitation with the estimated time series of phase changes. The graph of precipitation accumulation and bias time series is presented in Fig. 9(c). The correlation between the bias time series and cumulative precipitation indicates that the temporally inconsistent process may reflect the drying process in the soil after precipitation [59]. In general, the findings show a phase change with an increase in accumulation, especially in the wet period around June–September, accompanied by a phase change response to an increase in precipitation. Therefore, the impact of the increased precipitation on the phase difference was not instantly recorded. The lack of instantaneous phase jumps may be due to the inherent relative nature of the InSAR phase measurements with respect to the reference point. Meanwhile, in the November–January period, the accumulation of precipitation tends to stagnate, whereas phase changes also tend to stagnate.

However, the findings in this result indicate a significant phase change when the earthquake occurred, especially in November 2017, marked with a green line [see Fig. 9(c)]. During this period, there were phase differences related to earthquake events, but there was no increase in precipitation. In addition to the relationship with the earthquake, the liquefaction phenomenon could be one of the reasons behind the phase change in November 2017, even though there was no increase in precipitation. The phenomenon of liquefaction that affects soil moisture characteristics can provide differences in soil conditions, which were originally rigid and then tend to be soft, which can be captured as phase differences. By referring to reports from local institutions that show an average liquefaction of 5 m, it can contribute to this phase change phenomenon [17]. This particular pixel underwent a slight temporal decorrelation, and the time-series correlation showed a clear impact of the precipitation event. Moreover, the correlation recovery time corresponds to the duration of the phase change after the precipitation event. This is in line with the reported recovery time correlation with the duration of the phase change after the precipitation event. The derived bias time series likely reflects the InSAR phase response (relative to the reference pixel) to the same drying process after deposition [60].

D. Liquefaction Susceptibility Analysis

The DCNN algorithm with the optimization algorithm was employed to generate the model from the liquefaction-related factor after Pohang earthquake. The final model generated will be used as the liquefaction susceptibility map after reclassification into five classes using the natural (Jenks) method for each model, as shown in Fig. 10. Susceptibility maps will be divided into very low, low, moderate, high, and very high classes to identify the area susceptible to liquefaction. Then, the distribution of classes will be evaluated; from the result, the distribution of the result was noticed that the very low-to-low distribution was dominated in this area. High-very high-class distribution was clustered in certain spots over the area of interest. In addition, based on the geological distribution, this area has a lot of quaternary alluvium composition, which is

a type of soil that is quite susceptible to liquefaction, which is correlated with the tolerance of soil estrangement [61]. The distribution of blue areas indicating low liquefaction potential is found in areas with many settlements and high elevation, and although the liquefaction potential is quite low, the effects of damage caused by structures still accompany this area. Therefore, the prediction reliability of deep learning can vary depending on many variables, such as the proper preprocess data, quality, and quantity of datasets, and the model structure used in the learning process.

TABLE III
SPATIAL DETECTION ACCURACY ACCORDING TO SATELLITE AND INDEX

Spatial detection accuracy (%)				
	Landsat-8		Sentinel-2	
Threshold	NDWI Difference	TDL Index	NDWI Difference	TDL Index
0.000	23.56	74.31	61.89	92.37
0.025	12.43	54.32	28.36	56.71
0.05	2.31	11.78	2.47	27.62
0.1	0	3.12	0.78	10.65

V. DISCUSSION

This study used SAR data to detect liquefaction phenomena in Korea after the 2017 Pohang earthquake. Rice paddy areas with soft soil contain large amounts of moisture and are susceptible to vibration amplification, which leads to liquefaction. The liquefaction was related to changes in soil moisture conditions in the area around the earthquake epicenter. In addition to using the interferogram to analyze the liquefaction observations, we also used Sentinel-1 data to establish the coherence conditions in Pohang during 2017–2020. Changes in incoherence can shed light on changes in surface conditions, especially in areas with particular land cover types [54]. There was a relationship between the precipitation amounts and coherence conditions in the study area, especially in agricultural areas and those covered by paddy fields. However, there were some shortcomings in the analysis of changes in precipitation conditions and coherence values. Additionally, local deformation could lead to subsidence related to the post-earthquake liquefaction phenomenon in the paddy field area, which is composed of quaternary alluvium deposits. Local deformation occurred where the soil grains experienced changes in density due to a loss of pressure and increased pore water pressure in the displacement location. In addition, the rice paddy area had a coherence that was not excessively high, and the uncleared signal (i.e., interferogram pattern) could be related to the temporal decorrelation between the two data acquisition times. The integration between SAR analysis and optical analysis in detecting soil moisture changes has room to improve for separating clearly the signal related to changes in soil moisture. The nonsignificance closure phase may related to the multilook and filter used in this research during the process that affected the results [58]. Also, the area of liquefaction related to the farmland, which has seasonal conditions, affected the coherence and surface displacement conditions during time observation.

A. Validation of Optical Imagery Assessment

The validation process compared the image analysis results based on optical satellites and radar with existing field data. After the 2017 Pohang earthquake, observations of the liquefaction phenomenon were made, revealing more than 600 points of liquefaction [62], [63]. We determined the detection accuracy based on the number of pixels larger than the threshold value (taken as the affected area) and compared it with the field data. Table III shows the accuracy of data derived from optical satellites and radar applied in the analysis of areas affected by liquefaction. For Landsat-8, the NDWI difference value was 23.56%, while for Sentinel-2 (using the same method), the value was 61.89% at a threshold value of 0. The TDLI approach had values of 74.31% for Landsat-8 (threshold = 0) and 93.27% for Sentinel-2 (threshold = 0). In addition, the local deformation detected by InSAR had 72.45% concordance with the liquefaction data. The difference in detection rate may be attributable to the temporal conditions and effects of atmospheric disturbances during the interferogram generation process.

Based on published reports and a spatial assessment, the post-earthquake area of effect in Pohang had a diameter of 0–22 m, with an average sand bowl length of 5 m [63]. The resolution of Landsat-8 (∼30 m) may influence the detection quality, which was a factor in the low liquefaction index. The index values obtained using Sentinel-2 data were more useful for identifying areas of liquefaction due to the high resolution of Sentinel-2 (20 m at the subpixel level). In addition, temporal decorrelation, i.e., analysis two days after the earthquake, affected the water content post-liquefaction [10]. Landsat-8, which has longer intervals and greater coverage for the assessment of coseismic events, also allows liquefaction signals to be observed. We found that the results for both satellites aligned with liquefaction field data obtained by local institutions. This represents a promising approach to liquefaction analysis based on SAR data, enabling a better understanding of the phenomenon and providing a reference for future studies, especially in Pohang, South Korea.

B. Liquefaction Susceptibility Maps Performance Analysis

A liquefaction susceptibility map has been generated using the DCNN algorithm combined with remote sensing imagery to assess the liquefaction after Pohang earthquake. The CNN algorithm has provided a better advantage in the prediction task by generating a pattern that has a spatial relationship with the liquefaction. As for the validation result using k-fold cross validation, 30% of the inventory dataset designated as the test data has been used to assess the AUC value to evaluate the performance of the liquefaction susceptibility model [64]. For the performance analysis, the cross-validation k-fold was initiated to estimate the reliability of model performance. Cross validation splits the training into a number of folds of the same size and performs the training model in folds-1 [26], [65].

TABLE IV
COMPARISON OF PERFORMANCE EVALUATION FOR SUSCEPTIBILITY MAP ALGORITHM

	AUC	MSE	RMSE	MAE
Conventional CNN	0.76 ± 0.039	0.19 ± 0.029	0.38 ± 0.059	0.32 ± 0.025
DCNN	**0.81 ± 0.030**	**0.15 ± 0.026**	**0.30 ± 0.050**	**0.28 ± 0.022**
ResNet	0.77 ± 0.043	0.27 ± 0.037	0.55 ± 0.126	0.42 ± 0.022
UNet	0.78 ± 0.033	0.18 ± 0.031	0.37 ± 0.062	0.27 ± 0.032
LSTM	0.74 ± 0.026	0.24 ± 0.037	0.47 ± 0.075	0.38 ± 0.046

TABLE V
PERFORMANCE METRICS OF MODEL ALGORITHMS FOR LIQUEFACTIONS SUSCEPTIBILITY MAP USING DCNN AND DCNN-HYBRID METHODS

	DCNN	DCNN-GWO	DCNN-PSO
AUC	0.81	0.83	0.86
MSE	0.15	0.14	0.12
RMSE	0.30	0.27	0.24

The remaining folds are used for the validation phase in this analysis.

In order to validate the susceptibility model with DCNN architecture against the current state, we compare its performance with the conventional CNN, ResNet, UNet, and LSTM model using several metrics based on fivefold cross validation, as shown in Fig. 11. Our model (DCNN architecture) achieves the highest ROC-AUC of 0.81 ± 0.030 compared to another model. Furthermore, the DCNN model shows better performance compared to the different models, indicated by the lowest value with mse of (0.15 ± 0.026) and RMSE (0.30 ± 0.050), indicating more accurate predictions with less deviation from the ground truth values. The UNet, ResNet, and CNN architectures show moderate performance, with ROC-AUC scores of 0.78 ± 0.033, 0.77 ± 0.043, and 0.76 ± 0.040, respectively, while the LSTM model shows the lowest ROC-AUC performance (0.74 ± 0.026). The relatively small standard deviations across k-folds for most metrics indicate stable model performance, with DCNN showing consistent results across different data splits. Detailed performance evaluation results can be seen in Table IV. These findings suggest that the dense connectivity patterns and feature reuse in the DCNN architecture may be well suited for this particular application, possibly due to its ability to maintain and propagate diverse feature representations throughout the network. Thus, we continue to develop the hybrid model with the DCNN model as its base. These comprehensive results support the advancement of our proposed model compared to the current state-of-the-art methods.

The AUC evaluation shows a promising result in the performance of the liquefaction model from DCNN-base, DCNN-GWO, and DCNN-PSO with AUC 0.81, 0.83, and 0.86, respectively, as shown in Fig. 12(a)–(c). Moreover, the utilization of optimization algorithms shows improvement in hybrid algorithms with DCNN for susceptibility mapping.

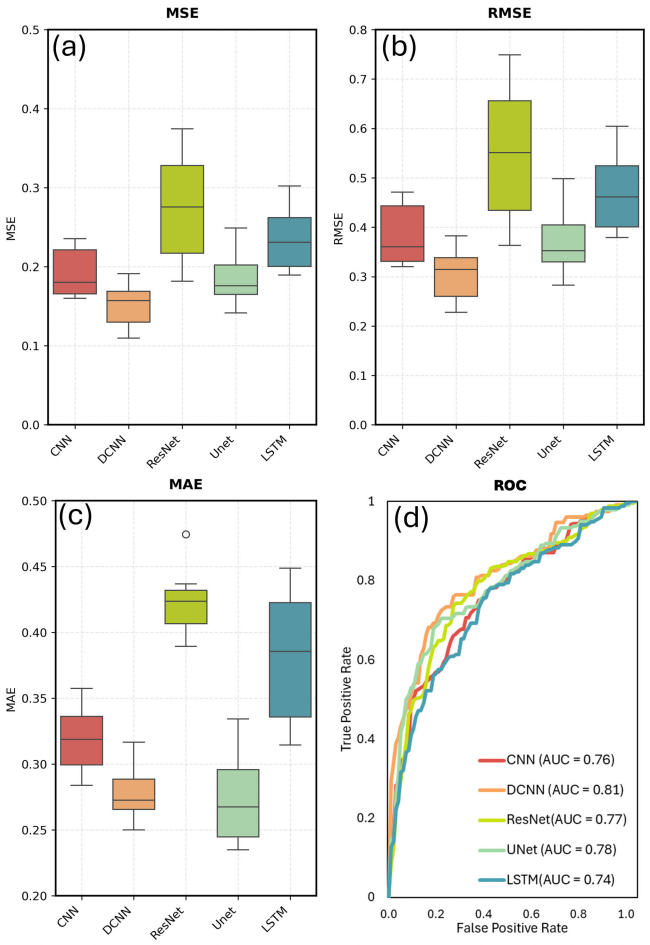


Fig. 11. Performance metric evaluation compared with DCNN architecture in (a) mse, (b) RMSE, (c) MAE, and (d) ROC analysis.

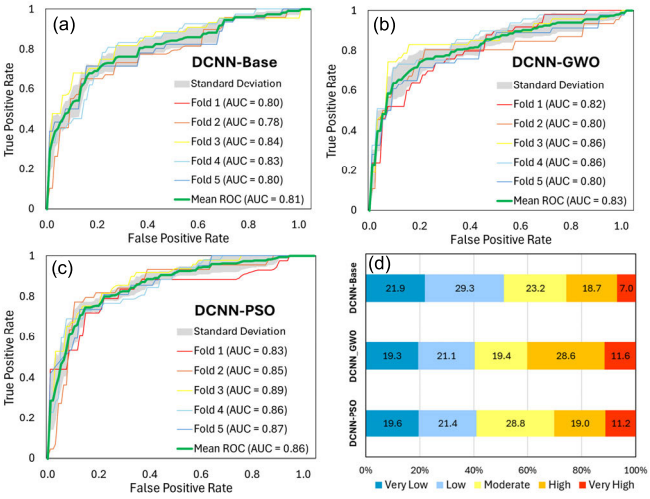


Fig. 12. ROC of the evaluated result from the liquefaction susceptibility map from (a) DCNN-base, (b) DCNN-GWO, (c) DCNN-PSO, and (d) distribution classes from different algorithms.

Based on the mse metric result, the DCNN, DCNN-GWO, and DCNN-PSO are shown as 0.15, 0.14, and 0.12, respectively, as shown in Table V. Then, for RMSE, the result from the hybrid model can be shown as 0.30 for DCNN, followed by 0.27 for DCNN-GWO and 0.24 for PSO; the lower value of mse and RMSE represents better performance of the model. Although the demonstration of the liquefaction prediction using the deep-learning model, there is still room for improvement that can be applied in the future. The distribution of training data and influential factor could affect the final result and needs to be considered during the training process of the model [66]. Besides, Fig. 12(d) shows the distribution of the susceptibility map for each class, with the moderate classes dominating this liquefaction susceptibility map for each model. Even so, this is the initial research in terms of a liquefaction prediction map using remote sensing data in Pohang, South Korea, and can be a reference for stakeholders in assessing the liquefaction to prevent loss in the future.

VI. Conclusion

A multidisciplinary remote sensing approach using both optical and radar data was conducted to correlate local changes reflected in DInSAR measurements with the observed liquefaction phenomenon. Liquefaction was investigated using SAR data; we quantitatively analyzed its spatial extent and the associated surface deformation. Surface deformation measured by DInSAR and surface changes revealed by SAR-related parameters (i.e., interference coherence and intensity correlations) enabled the identification of zones that may affected by liquefaction. A liquefaction analysis and calculation of liquefaction loss were conducted in the study area. The results of the bias time-series analysis conducted in the paddy plain area were in line with the observed changes in soil moisture conditions revealed by the optical satellite data. Due to water shortages in shallow soil layers and high vertical overload pressure in deep sandy soils, soils cannot be deformed by the liquefaction of shallow or deep sand layers.

The results of this study will enable the application of SAR data in analyses of post-earthquake phenomena with spatial analysis accuracy above 70% compared with field survey data. This study introduces the use of SAR data to analyze liquefaction in South Korea, especially after Pohang earthquake. The liquefaction susceptibility map has been generated by employing the DCNN algorithms using satellite imagery in Pohang area. The DCNN algorithm used in the susceptibility map exhibits promising performance with an AUC value of 0.81, 0.83, and 0.86 for DCNN-base, DCNN-GWO, and DCNN-PSO, respectively. Then, the algorithm shows the advantages of managing and analyzing the spatial and nonlinear data from satellite imagery. Also, this study reveals the usefulness of a machine-learning algorithm to evaluate the nonlinear problem in geospatial problems. In addition, decomposition analysis based on SAR data (amplitude, interferometry, and coherence) and optical imagery can provide comprehensive analysis. Moreover, this research could also increase awareness of liquefaction and the utility of SAR data for mitigating and mapping liquefaction disasters.

References

[1] L. Chen et al., "Liquefaction macrophenomena in the great wenchuan earthquake," *Earthq. Eng. Eng. Vib.*, vol. 8, no. 2, pp. 219–229, Jun. 2009, doi: 10.1007/s11803-009-9033-4.

[2] A. Oro et al., "Surface deformation and secondary effects of the January 18, 2021 (Mw 6.5) San Juan (Argentina) earthquake from remote sensing techniques," *Int. J. Earth Sci.*, vol. 112, no. 8, pp. 2267–2291, Nov. 2023, doi: 10.1007/s00531-023-02354-x.

[3] X. Bao, Z. Jin, H. Cui, X. Chen, and X. Xie, "Soil liquefaction mitigation in geotechnical engineering: An overview of recently developed methods," *Soil Dyn. Earthq. Eng.*, vol. 120, pp. 273–291, May 2019, doi: 10.1016/j.soildyn.2019.01.020.

[4] Y. Huang and M. Yu, "Review of soil liquefaction characteristics during major earthquakes of the twenty-first century," *Natural Hazards*, vol. 65, no. 3, pp. 2375–2384, Oct. 2012, doi: 10.1007/s11069-012-0433-9.

[5] A. S. Nur and C. W. Lee, "Damage proxy map (DPM) of the 2016 Gyeongju and 2017 Pohang earthquakes using Sentinel-1 imagery," *Korean J. Remote Sens.*, vol. 37, no. 1, pp. 13–22, 2021, doi: 10.7780/kjrs.2021.37.1.2.

[6] H.-S. Kim, C.-G. Sun, and H.-I. Cho, "Geospatial assessment of the post-earthquake hazard of the 2017 Pohang earthquake considering seismic site effects," *ISPRS Int. J. Geo-Inf.*, vol. 7, no. 9, p. 375, Sep. 2018, doi: 10.3390/ijgi7090375.

[7] R. Bamler and P. Hartl, "Synthetic aperture radar interferometry," *Inverse Problems*, vol. 14, no. 4, pp. 1–54, Aug. 1998, doi: 10.1088/0266-5611/14/4/001.

[8] W. L. Hakim, M. F. Fadhillah, S. Park, B. Pradhan, J.-S. Won, and C.-W. Lee, "InSAR time-series analysis and susceptibility mapping for land subsidence in semarang, Indonesia using convolutional neural network and support vector regression," *Remote Sens. Environ.*, vol. 287, Mar. 2023, Art. no. 113453, doi: 10.1016/j.rse.2023.113453.

[9] M. F. Fadhillah et al., "Surface deformation simulation for InSAR detection using a machine learning approach on the hantangang river volcanic field: A case study on the orisan mountain," *Frontiers Environ. Sci.*, vol. 10, p. 1412, Aug. 2022, doi: 10.3389/fenvs.2022.968120.

[10] H.-S. Kim, M. Kim, L. G. Baise, and B. Kim, "Local and regional evaluation of liquefaction potential index and liquefaction severity number for liquefaction-induced sand boils in Pohang, South Korea," *Soil Dyn. Earthq. Eng.*, vol. 141, Feb. 2021, Art. no. 106459, doi: 10.1016/j.soildyn.2020.106459.

[11] E. Bol et al., "Evaluation of soil liquefaction in the city of Hatay triggered after the February 6, 2023 Kahramanmaraş-Türkiye earthquake sequence," *Eng. Geol.*, vol. 339, Sep. 2024, Art. no. 107648, doi: 10.1016/j.enggeo.2024.107648.

[12] M. Chini et al., "Coseismic liquefaction phenomenon analysis by COSMO-SkyMed: 2012 Emilia (Italy) earthquake," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 39, pp. 65–78, Jul. 2015, doi: 10.1016/j.jag.2015.02.008.

[13] K. Ishitsuka, T. Tsuji, and T. Matsuoka, "Detection and mapping of soil liquefaction in the 2011 Tohoku earthquake using SAR interferometry," *Earth, Planets Space*, vol. 64, no. 12, pp. 1267–1276, Jan. 2013, doi: 10.5047/eps.2012.11.002.

[14] Q. Li, W. Wang, J. Wang, J. Zhang, and D. Geng, "Exploring the relationship between InSAR coseismic deformation and earthquake-damaged buildings," *Remote Sens. Environ.*, vol. 262, Sep. 2021, Art. no. 112508, doi: 10.1016/j.rse.2021.112508.

[15] Y. Eshqi Molan and Z. Lu, "Modeling InSAR phase and SAR intensity changes induced by soil moisture," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4967–4975, Jul. 2020, doi: 10.1109/TGRS.2020.2970841.

[16] O. L. Stephenson et al., "Deep learning-based damage mapping with InSAR coherence time series," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5207917, doi: 10.1109/TGRS.2021.3084209.

[17] H. Baik, Y.-S. Son, and K.-E. Kim, "Detection of liquefaction phenomena from the 2017 Pohang (Korea) earthquake using remote sensing data," *Remote Sens.*, vol. 11, no. 18, p. 2184, Sep. 2019, doi: 10.3390/rs11182184.

[18] A. Mokhtari, M. Sadeghi, Y. Afrasiabian, and K. Yu, "OPTRAM-ET: A novel approach to remote sensing of actual evapotranspiration applied to Sentinel-2 and Landsat-8 observations," *Remote Sens. Environ.*, vol. 286, Mar. 2023, Art. no. 113443, doi: 10.1016/j.rse.2022.113443.

[19] N. Mzid, R. Casa, S. Pascucci, M. Tolomio, and S. Pignatti, "Assessment of the potential of PRISMA hyperspectral data to estimate soil moisture," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2022, pp. 5606–5609, doi: 10.1109/IGARSS46834.2022.9883716.

[20] Y. Zhong, Z. Wei, L. Miao, Y. Wang, J. P. Walker, and A. Colliander, "Downscaling passive microwave soil moisture estimates using standalone optical remote sensing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4504919, doi: 10.1109/TGRS.2024.3386967.

[21] S. G. Song and H. Lee, "Static slip model of the 2017 M_w 5.4 Pohang, South Korea, earthquake constrained by the InSAR data," *Seismol. Res. Lett.*, vol. 90, no. 1, pp. 140–148, Feb. 2019, doi: 10.1785/0220180156.

[22] Y. Wang, S. Shu, and Y. Wu, "Reliability analysis of soil liquefaction considering spatial variability of soil property," *J. Earthq. Tsunami*, vol. 16, no. 1, 2021, Art. no. 2250002, doi: 10.1142/S1793431122500026.

[23] A. Hrysiewicz, E. P. Holohan, S. Donohue, and H. Cushnan, "SAR and InSAR data linked to soil moisture changes on a temperate raised peatland subjected to a wildfire," *Remote Sens. Environ.*, vol. 291, Jun. 2023, Art. no. 113516, doi: 10.1016/j.rse.2023.113516.

[24] H. Ansari, F. De Zan, and A. Parizzi, "Study of systematic bias in measuring surface deformation with SAR interferometry," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1285–1301, Feb. 2021, doi: 10.1109/TGRS.2020.3003421.

[25] Y. E. Molan, Z. Lu, and J.-W. Kim, "Influence of the statistical properties of phase and intensity on closure phase," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7346–7354, Oct. 2020, doi: 10.1109/TGRS.2020.2982062.

[26] L. Lv, T. Chen, G. Liu, J. Dou, and A. Plaza, "A comparative study of model interpretability considering the decision differentiation of landslide susceptibility models," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2024, Art. no. 4401218, doi: 10.1109/TGRS.2024.3522165.

[27] H. J. Oh and S. Lee, "Integration of ground subsidence hazard maps of abandoned coal mines in Samcheok, Korea," *Int. J. Coal Geol.*, vol. 86, no. 1, pp. 58–72, Apr. 2011, doi: 10.1016/j.coal.2010.11.009.

[28] S. Lee, I. Park, and J. K. Choi, "Spatial prediction of ground subsidence susceptibility using an artificial neural network," *Environ. Manage.*, vol. 49, no. 2, pp. 347–358, 2012, doi: 10.1007/s00267-011-9766-5.

[29] S. Lee and C.-W. Lee, "Application of decision-tree model to groundwater productivity-potential mapping," *Sustainability*, vol. 7, no. 10, pp. 13416–13432, Sep. 2015, doi: 10.3390/su71013416.

[30] M. F. Fadhillah, A. R. Achmad, and C.-W. Lee, "Improved combined scatterers interferometry with optimized point scatterers (ICOPS) for interferometric synthetic aperture radar (InSAR) time-series analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5220014, doi: 10.1109/TGRS.2021.3138763.

[31] C. W. Lee, Z. Lu, J. S. Won, H. S. Jung, and D. Dzurisin, "Dynamic deformation of Seguam Island, Alaska, 1992–2008, from multi-interferogram InSAR processing," *J. Volcanol. Geothermal Res.*, vol. 260, pp. 43–51, Jun. 2013, doi: 10.1016/j.jvolgeores.2013.05.009.

[32] A. Parizzi and R. Brcic, "Adaptive InSAR stack multilooking exploiting amplitude statistics: A comparison between different techniques and practical results," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 3, pp. 441–445, May 2011, doi: 10.1109/LGRS.2010.2083631.

[33] F. Zhao and J. J. Mallorqui, "A temporal phase coherence estimation algorithm and its application on DInSAR pixel selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8350–8361, Nov. 2019, doi: 10.1109/TGRS.2019.2920536.

[34] A. Hooper, D. Bekaert, K. Spaans, and M. Arikan, "Recent advances in SAR interferometry time series analysis for measuring crustal deformation," *Tectonophysics*, vols. 514–517, pp. 1–13, Jan. 2012, doi: 10.1016/j.tecto.2011.10.013.

[35] M. Jiang, X. Ding, R. F. Hanssen, R. Malhotra, and L. Chang, "Fast statistically homogeneous pixel selection for covariance matrix estimation for multitemporal InSAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1213–1224, Mar. 2015, doi: 10.1109/TGRS.2014.2336237.

[36] X. Wang, Y. Aoki, and J. Chen, "Surface deformation of Asama volcano, Japan, detected by time series InSAR combining persistent and distributed scatterers, 2014–2018," *Earth, Planets Space*, vol. 71, no. 1, p. 121, Dec. 2019, doi: 10.1186/s40623-019-1104-9.

[37] M. F. Fadhillah et al., "Multitemporal analysis of land subsidence induced by open-pit mining activity using improved combined scatterer interferometry with deep learning algorithm optimization," *Science Rep.*, vol. 14, no. 1, p. 6311, Mar. 2024, doi: 10.1038/s41598-024-56347-0.

[38] C. W. Lee, Z. Lu, and H. S. Jung, "Simulation of time-series surface deformation to validate a multi-interferogram InSAR processing technique," *Int. J. Remote Sens.*, vol. 33, no. 22, pp. 7075–7087, 2012, doi: 10.1080/01431161.2012.700137.

[39] P. Lu, S. Bai, V. Tofani, and N. Casagli, "Landslides detection through optimized hot spot analysis on persistent scatterers and distributed scatterers," *ISPRS J. Photogramm. Remote Sens.*, vol. 156, pp. 147–159, Oct. 2019, doi: 10.1016/j.isprsjprs.2019.08.004.

[40] J. K. Ord and A. Getis, "Local spatial autocorrelation statistics: Distributional issues and an application," *Geographical Anal.*, vol. 27, no. 4, pp. 286–306, Oct. 1995, doi: 10.1111/j.1538-4632.1995.tb00912.x.

[41] B. C. Gao, "NDWI-A normalized difference water index for remote sensing of vegetation liquid water from space," *Remote Sens. Environ.*, vol. 58, no. 3, pp. 257–266, Dec. 1996, doi: 10.1016/S0034-4257(96)00067-3.

[42] D. B. Lobell and G. P. Asner, "Moisture effects on soil reflectance," *Soil Sci. Soc. Amer. J.*, vol. 66, no. 3, pp. 722–727, May 2002, doi: 10.2136/sssaj2002.7220.

[43] M. Sadeghi, N. Mohamadzadeh, L. Liang, U. Bandara, M. M. Caldas, and T. Hatch, "A new variant of the optical trapezoid model (OPTRAM) for remote sensing of soil moisture and water bodies," *Sci. Remote Sens.*, vol. 8, Dec. 2023, Art. no. 100105, doi: 10.1016/j.srs.2023.100105.

[44] E. Babaeian, M. Sadeghi, T. E. Franz, S. Jones, and M. Tuller, "Mapping soil moisture with the optical trapezoid model (OPTRAM) based on long-term MODIS observations," *Remote Sens. Environ.*, vol. 211, pp. 425–440, Jun. 2018, doi: 10.1016/j.rse.2018.04.029.

[45] M. H. Vahidnia, A. A. Alesheikh, A. Alimohammadi, and F. Hosseinali, "A GIS-based neuro-fuzzy procedure for integrating knowledge and data in landslide susceptibility mapping," *Comput. Geosci.*, vol. 36, no. 9, pp. 1101–1114, Sep. 2010, doi: 10.1016/j.cageo.2010.04.004.

[46] W. L. Hakim, M. F. Fadhillah, K.-J. Lee, S.-J. Lee, S.-H. Chae, and C.-W. Lee, "Land subsidence and groundwater storage assessment using ICOPS, GRACE, and susceptibility mapping in Pekalongan, Indonesia," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5218225, doi: 10.1109/TGRS.2023.3324043.

[47] P. T. T. Ngo et al., "Evaluation of deep learning algorithms for national scale landslide susceptibility mapping of Iran," *Geosci. Frontiers*, vol. 12, no. 2, pp. 505–519, Mar. 2021, doi: 10.1016/j.gsf.2020.06.013.

[48] H. S. Elsayed, O. M. Saad, M. S. Soliman, Y. Chen, and H. A. Youness, "Attention-based fully convolutional DenseNet for earthquake detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5918610, doi: 10.1109/TGRS.2022.3194196.

[49] C. Zhang, G. Li, and S. Du, "Multi-scale dense networks for hyperspectral remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 9201–9222, Nov. 2019, doi: 10.1109/TGRS.2019.2925615.

[50] M. F. Fadhillah, W. L. Hakim, M. Panahi, F. Rezaie, C.-W. Lee, and S. Lee, "Mapping of landslide potential in Pyeongchang-gun, South Korea, using machine learning meta-based optimization algorithms," *Egyptian J. Remote Sens. Space Sci.*, vol. 25, no. 2, pp. 463–472, Aug. 2022, doi: 10.1016/j.ejrs.2022.03.008.

[51] W. L. Hakim et al., "Convolutional neural network (CNN) with metaheuristic optimization algorithms for landslide susceptibility mapping in Icheon, South Korea," *J. Environ. Manage.*, vol. 305, Mar. 2022, Art. no. 114367, doi: 10.1016/j.jenvman.2021.114367.

[52] A.-L. Balogun et al., "Spatial prediction of landslide susceptibility in western Serbia using hybrid support vector regression (SVR) with GWO, BAT and COA algorithms," *Geosci. Frontiers*, vol. 12, no. 3, May 2021, Art. no. 101104, doi: 10.1016/j.gsf.2020.10.009.

[53] F. A. Senel, F. Gökçe, A. S. Yüksel, T. Yiğit, "A novel hybrid PSO–GWO algorithm for optimization problems," *Eng. Comput.*, vol. 35, no. 4, pp. 1359–1373, Oct. 2019, doi: 10.1007/s00366-018-0668-5.

[54] P. M. Bürgi and R. B. Lohman, "High-resolution soil moisture evolution in hyper-arid regions: A comparison of InSAR, SAR, microwave, optical, and data assimilation systems in the Southern Arabian Peninsula," *J. Geophys. Res., Earth Surf.*, vol. 126, no. 12, Dec. 2021, Art. no. e2021JF006158, doi: 10.1029/2021JF006158.

[55] S. Zwieback, S. Hensley, and I. Hajnsek, "Soil moisture estimation using differential radar interferometry: Toward separating soil moisture and displacements," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 9, pp. 5069–5083, Sep. 2017, doi: 10.1109/TGRS.2017.2702099.

[56] K. V. Singh, R. Setia, S. Sahoo, A. Prasad, and B. Pateriya, "Evaluation of NDWI and MNDWI for assessment of waterlogging by integrating digital elevation model and groundwater level," *Geocarto Int.*, vol. 30, no. 6, pp. 650–661, Jul. 2015, doi: 10.1080/10106049.2014.965757.

[57] F. De Zan and G. Gomba, "Vegetation and soil moisture inversion from SAR closure phases: First experiments and results," *Remote Sens. Environ.*, vol. 217, pp. 562–572, Nov. 2018, doi: 10.1016/j.rse.2018.08.034.

[58] Y. Zheng, H. Fattahi, P. Agram, M. Simons, and P. Rosen, "On closure phase and systematic bias in multilooked SAR interferometry," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5226611, doi: 10.1109/TGRS.2022.3167648.

[59] F. De Zan, M. Zonno, and P. López-Dekker, "Phase inconsistencies and multiple scattering in SAR interferometry," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 12, pp. 6608–6616, Dec. 2015, doi: 10.1109/TGRS.2015.2444431.

[60] T. E. Jordan et al., "Surface materials and landforms as controls on InSAR permanent and transient responses to precipitation events in a hyperarid desert, Chile," *Remote Sens. Environ.*, vol. 237, Feb. 2020, Art. no. 111544, doi: 10.1016/j.rse.2019.111544.

[61] Y. S. Gihm et al., "Paleoseismological implications of liquefaction-induced structures caused by the 2017 Pohang earthquake," *Geosci. J.*, vol. 22, no. 6, pp. 871–880, Dec. 2018, doi: 10.1007/s12303-018-0051-y.

[62] J. U. Woo et al., "An in-depth seismological analysis revealing a causal link between the 2017 M_W 5.5 Pohang earthquake and EGS project," *J. Geophys. Res., Solid Earth*, vol. 124, no. 12, pp. 13060–13078, Dec. 2019, doi: 10.1029/2019JB018368.

[63] K. H. Kim, J. H. Ree, Y. H. Kim, S. Kim, S. Y. Kang, and W. Seo, "Assessing whether the 2017 M_w 5.4 Pohang earthquake in South Korea was an induced event," *Science*, vol. 360, no. 6392, pp. 1007–1009, Jun. 2018, doi: 10.1126/science.aat6081.

[64] A. Arabameri et al., "A novel ensemble computational intelligence approach for the spatial prediction of land subsidence susceptibility," *Sci. Total Environ.*, vol. 726, Jul. 2020, Art. no. 138595, doi: 10.1016/j.scitotenv.2020.138595.

[65] T.-T. Wong and P.-Y. Yeh, "Reliable accuracy estimates from k-fold cross validation," *IEEE Trans. Knowl. Data Eng.*, vol. 32, no. 8, pp. 1586–1594, Aug. 2020, doi: 10.1109/TKDE.2019.2912815.

[66] N. Anantrasirichai, J. Biggs, F. Albino, and D. Bull, "The application of convolutional neural networks to detect slow, sustained deformation in InSAR time series," *Geophys. Res. Lett.*, vol. 46, no. 21, pp. 11850–11858, Nov. 2019, doi: 10.1029/2019GL084993.



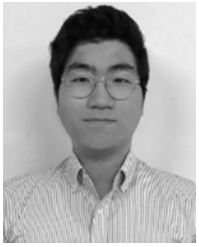
Muhammad Fulki Fadhillah was born in Tasikmalaya, West Java, Indonesia, in 1996. He received the B.S. degree in physics from Indonesia University of Education, Bandung, Indonesia, in 2019. He is currently pursuing the combined M.Sc. and Ph.D. degree in integrated science with Kangwon National University, Chuncheon-si, South Korea.

From 2017 to 2018, he was a Research Assistant with the Basic Physics Laboratory, Department of Physics Education, Indonesia University of Education. Since 2020, he has been a Research Assistant with Kangwon National University Earth Observation Laboratory, Kangwon National University. His research interests include synthetic aperture radar (SAR) interferometry, time-series interferometry SAR (InSAR), geographic information systems (GISs), and machine learning.



Wahyu Luqmanul Hakim received the B.Sc. degree in physics science from Indonesia University of Education, Bandung, Indonesia, in 2019, and the Doctor of Philosophy degree in natural science from the Department of Science Education, majoring in Integrative Science.

He completed a combined M.Sc. and Ph.D. program in 2025. He is currently a Postdoctoral Researcher with the Department of Science Education, Kangwon National University, and the Interdisciplinary Program in Earth Environmental System Science and Engineering, South Korea. His research focuses on remote sensing and geographic information systems (GIS), particularly synthetic aperture radar (SAR) interferometry, time-series InSAR, GIS analysis, and susceptibility mapping.



Sung-Jae Park received the B.S. and M.S. degrees from Kangwon National University, Chuncheon-si, South Korea, in 2017 and 2019, respectively, where he is currently pursuing the Ph.D. degree with the Department of Smart Regional Innovation.

He is currently a member with Kangwon National University Earth Observation Laboratory (KEOL), Kangwon National University. His research interests focus on remote sensing and geographic information system (GIS) applications.



Chang-Wook Lee (Senior Member, IEEE) received the B.S. degree from Kangwon National University, Chuncheon-si, South Korea, in 2000, and the M.S. and Ph.D. degrees in earth system sciences (remote sensing) from Yonsei University, Seoul, South Korea, in 2002 and 2009, respectively.

He had filled the postdoctoral position in InSAR for the ARTS contract with the U.S. Geological Survey EROS Data Center and all work was to be performed as determined by the USGS project Manager working in coordination with Science Support Services Contract (SSSC) Management. The work was performed at the USGS Cascades Volcano Observatory in Vancouver, Washington, by National Aeronautics and Space Administration (NASA) Project support from 2009 to 2011. He is a Professor with the Department of Science Education and Department of Smart Regional Innovation, Kangwon National University. His research interests include SAR, InSAR, and timeseries processing technique development on natural disaster monitoring and resource characterization. He has authored more than 100 articles in these research fields.